

BALAMURUGAN C

+91 91598 50002 | tharanishbalaa@gmail.com | Coimbatore, Tamil Nadu | linkedin.com/in/balamurugan-c-5507b82a3 | github.com/Balamurugan0113 | bala-portfolio-sigma.vercel.app

PROFESSIONAL SUMMARY

Results-driven Final Year B.Tech AI & Data Science student and Python Developer specializing in Artificial Intelligence, Machine Learning, Data Science, and Full Stack Web Development. Hands-on experience developing real-world applications with FastAPI, React, and PostgreSQL. Skilled in data preprocessing, analytical modeling, and intelligent web applications. Proactive problem solver dedicated to engineering scalable AI solutions, computer vision models, and robust full-stack architectures.

TECHNICAL SKILLS

- **Programming Languages:** Python, JavaScript, TypeScript, SQL
- **AI & Machine Learning:** TensorFlow, Scikit-Learn, Pandas, NumPy, OpenCV, Generative AI
- **Web Development:** React, FastAPI, HTML5, CSS3, RESTful APIs, Node.js
- **Databases:** PostgreSQL, MySQL
- **Developer Tools:** Git, GitHub, VS Code, Docker, Oracle Cloud Infrastructure (OCI)
- **Operating Systems:** Windows, Linux (Kali Linux)

WORK EXPERIENCE & INTERNSHIP

Python with Data Science Intern | ACCENT TECHNO SOFT

- Architected end-to-end Python scripts for automated data preprocessing, exploratory data analysis (EDA), and feature extraction on structured datasets.
- Implemented supervised machine learning classification and regression models using Scikit-Learn and NumPy, improving predictive accuracy.
- Created interactive data visualizations using Matplotlib and Pandas to derive actionable analytical insights and business intelligence.
- Collaborated on real-world AI project prototypes, applying best practices in model evaluation, hyperparameter tuning, and workflow optimization.

FEATURED PROJECTS

INTELLIQ – AI & DS Department Association Web Platform | Full Stack Web Application

Technologies: React, FastAPI, PostgreSQL, HTML5/CSS3, REST APIs

- Designed and engineered the official full-stack web application for the AI & Data Science Department Association at Info Institute of Engineering.
- Developed interactive portals for departmental event management, student project showcases, member registrations, and real-time notifications.
- Optimized backend RESTful API endpoints and database queries to ensure smooth user experience and high data availability.

Campus360 – AI-Powered College ERP Platform | Full Stack & AI Solution

Technologies: React, FastAPI, PostgreSQL, Python, REST APIs

- Designed and built an enterprise-grade college ERP solution featuring Student Portal, Faculty Portal, Bus Tracking, and Attendance modules.
- Integrated an intelligent AI Chatbot backend with FastAPI to handle real-time student queries, notifications, and automated query resolution.
- Optimized database schema and API endpoints in PostgreSQL to deliver high-concurrency response times and seamless UI updates in React.

Handwritten Digit Recognition System | Deep Learning / Computer Vision

Technologies: Python, TensorFlow, Keras, NumPy, Matplotlib, OpenCV

- Engineered a Convolutional Neural Network (CNN) trained on the MNIST dataset to classify handwritten digits with high precision (>97% accuracy).
- Implemented image preprocessing pipelines including grayscale conversion, thresholding, and matrix normalization to process raw visual inputs.
- Built visual performance metrics and confusion matrices using Matplotlib to evaluate model performance across varied image samples.

COMPETITIONS & HACKATHONS

- Smart India Hackathon (SIH 2025) – Participant in National-Level Innovation Hackathon
- Trisquadathon 2025 – Participant in Campus Hackathon, Info Institute of Engineering

CERTIFICATIONS

- **Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate**
Issued by: Oracle University | Issued: 2025
- **Oracle Cloud Infrastructure 2025 Certified Foundations Associate**
Issued by: Oracle University | Issued: 2025
- **Generative AI in Action**
Issued by: IBM SkillsBuild | Issued: 2025
- **Foundations of Cybersecurity**
Issued by: Google (via Coursera) | Issued: 2025

EDUCATION

B.Tech in Artificial Intelligence & Data Science (Final Year) | 2023 – 2027

Info Institute of Engineering | Kovilpalayam, Coimbatore

Secondary Schooling (11th – 12th) | 2021 – 2023

Shri Nehru Vidyalaya Matriculation Higher Secondary School | R.S. Puram, Coimbatore

KEY ACHIEVEMENTS & LANGUAGES

- Developed and deployed production-grade AI and Full Stack web applications with active GitHub codebases.
- Designed and maintained personal developer portfolio website showcasing interactive web applications.
- Languages: English, Tamil.

HOBBIES & INTERESTS

- **Handball Player:** Since school days
- **PC Gaming:** Valorant, GTA 5, Strategy games
- **Mobile Gaming:** Casual & competitive gaming
- **Freelancer:** Web and security project delivery
- **Final Year Projects:** Supporting college students with capstone and final year project work